

STA 137 Project

An Analysis of the Central African Republic's Export Time Series

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Background

The Central African Republic is a developing country that has undergone serious political and economic corruption since its inception in 1960. Originally a French colony, the Central African Republic is rich in natural resources, such as diamonds, timber, cotton, coffee, gold, and other precious metals. Despite the country having these resources, the infrastructure is extremely poor and lackluster, leading to most of the population working in agriculture to feed their families. Because of this, exports are one of the most significant contributors to the country's economy, thus making it dependent on international markets.

Our main goal of this analysis is to understand the Central African Republic's exports of goods as a percentage of its GDP, and any patterns or trends that may exist from 1960 to 2017. With this background knowledge, analyzing the time series of the exports allows us to discover more about what is happening inside the data and how time points are related to past events. The exports are highly susceptible to political and economic influences, so our modeling must account for variations in the contribution of exports to the GDP. By analyzing plots and performing transformations on the data, we can synthesize a model that adequately explains the Central African Republic's exports over the 57-year window. Beyond the end time point of the data, we are interested in understanding how the export percentages typically behave and what our model forecasts the export percentages to be in subsequent years.

Exploratory Data Analysis

At first glance at the raw time series (**Model 1.1**), there appears to be a downward trend in the data. This is further supported by our kernel smoothing (**Model 1.2**) where we can see that following the peak exports in 1968, our trend tends to curve downwards until the end of the time series. Furthermore, the raw time series indicates that this is a nonstationary time series with a nonconstant mean and unequal variance. Our kernel smoothing demonstrates that our lines do not center around a mean and the unequal growth and decrease in the export percentages indicate that the variance is not constant. This shows that we should explore a log transformation to fix the variance and first-order differencing in order to restore stationarity to the time series. Because this is an annual time series dataset, there is no purpose in exploring the seasonality of the exports data. There are also no significant outliers in our data.

The export percentage of GDP has a minimum of 10.68 and a maximum of 34.31, which is a large variation as represented in the plots. We obtain a mean of 20.66 and a median of 21.74 which are relatively close to one another, as well as a Fisher-Pearson skewness coefficient of .03. Since our measures of central tendency are so close together and are approximately the same distance from the maximum and minimum values, it makes sense to say that the export data is not skewed. Our skewness coefficient supports the lack of skew since it is approximately 0, permitting us to assume that our data is relatively symmetric around the mean. These statistics reveal to us that our dataset does not need transformations to address symmetry, so we only need to apply transformations intended to fix unequal variance and nonconstant mean.

Table 1: Descriptive Statistics

Minimum	Maximum	Mean	Median	Skewness
10.68	34.31	20.66	21.74	.03

Transformation

The ACF graph [Figure 1.1] of the raw shows a slow decay across lags, and the PACF graph [Figure 1.2] shows a significant spike at lag 1 and 2 and then gradually decreases and declines; both the ACF and PACF indicate the time series is not stationary. To address the non-stationarity indicated by the ACF and PACF patterns, we implemented a log transformation to stabilize the variance (**Model 2.1**), followed by differencing to remove trends, ensuring the data was suitable for time series modeling. As shown in [Figure 2.1 and 2.2], after log transformation, the ACF and PACF graphs look very similar to the raw data, with no meaningful information. However, after differencing the time series (**Model 3.1**), ACF graphs [Figure 3.1] show a positive spike at lag 3 and a negative spike at lag 1, indicating potential moving average (MA) components; PACF graph [Figure 3.2] has negative spikes at lags 1 and 2, which suggests that an autoregressive (AR) component may be necessary. Based on the ACF and PACF analysis, which suggests the presence of both autoregressive (AR) and moving average (MA) components, we come up with the following model candidates: ARIMA(1,1,0), ARIMA(2,1,0), ARIMA(0,1,2), ARIMA(0,1,3), ARIMA(2,1,2).

Model Selection

ARIMA(1,1,0) (**Model 4.1**), ARIMA(0,1,2)(**Model 5.1**) and ARIMA(0,1,3)(**Model 6.1**) have significant ACF of residual, indicating the assumption of white noise is not met, thus ARIMA(1,1,0), ARIMA(0,1,2) and ARIMA(0,1,3) are dropped.

For ARIMA (2,1,0) (**Model 7.1**) and ARIMA (2,1,2) (**Model 8.1**), The ACF residual plots showed no significant lag, which means the residuals resemble white noise. Additionally, the residuals normality plots suggested that the residuals are approximately normally distributed. Thus, these two models are strong candidates for our analysis.

For the ARIMA(2,1,0) / AR(2) (**Model 7.1**) model, the constant has p-value greater than 0.05, indicating the constant term does not significantly contribute to the model, thus it is dropped.

$$y_t = -0.8933 y_{t-1} - 0.06421 y_{t-2} + w_t$$

For ARIMA(2,1,2) / ARMA(2,2)(**Model 8.1**) model, the constant has p-value greater than 0.05, also indicating the constant term does not significantly contribute to the model, the MA(2) term also has p-value greater than 0.05, meaning statistically the moving average term at lag 2 may not be relevant for the model.

$$y'_t = -0.6926 y_{t-1} - 0.4081 y_{t-2} - 0.6712 w_{t-1} - 0.3288 w_{t-2} + w_t$$

As the moving average term at lag 2 appears to lack statistical relevance, the model is re-estimated as ARIMA(2, 1, 1) (**Model 9.1**) / ARMA(2,1). Again, the constant has p-value greater than 0.05, so the constant term is dropped.

$$y''_t = -0.3981 y_{t-1} - 0.3003 y_{t-2} - w_{t-1} + w_t$$

Table 2: Model Selection Criteria for ARIMA Models

	AIC	BIC
ARIMA(2,1,0)	-60.58510	-54.50905

ARIMA(2,1,2)	-67.52584	-57.39908
ARIMA(2,1,1)	-67.89777	-59.79637

ARIMA(2,1,2) and ARIMA(2,1,1) have much lower AIC and BIC scores than ARIMA(2,1,0), thus ARIMA(2,1,0) is eliminated; ARIMA(2,1,2) and ARIMA(2,1,1) have similar scores, but ARIMA(2,1,1) is slightly lower on both AIC BIC scores and it is a simpler model, thus ARIMA(2,1,1) is the final model.

Forecast

[Figure 9.1] forecasts with the best performing ARIMA(2,1,1) model applied to the log-differenced Central African Republic exports over time. Historical data from 1960 to 2017 displays notable volatility, with alternating peaks and troughs reflecting fluctuations in export growth rates. The forecasted period, shaded in blue, provides predictions along with confidence intervals to quantify uncertainty. The central forecast line indicates a relatively stable trend post-2020, which implies that significant upward or downward shifts in exports are unlikely under current conditions. However, the historical volatility and the widening confidence intervals, as the forecast extends further into the future, highlight the possibility of unpredictable shocks that could impact export performance. This suggests that while immediate export conditions might stabilize, external factors such as global trade dynamics, commodity price changes, or economic policies could significantly impact long-term performance. This visualization reflects the inherent challenges in predicting volatile time series, as evidenced by the widening confidence bands.

Conclusion

Through our exploration of the data, we are able to identify that the raw time series is a non-stationary time series with non-constant mean and unequal variance, which drove us to explore further through log transformation and differencing to achieve stationarity for our time series modeling. We only applied transformation to address the issues of unequal variance and non-constant mean, as the data showed symmetry and did not suggest a need to adjust skewness.

Based on the ACF and PACF analysis from our transformations, the presence of both autoregressive (AR) and moving average (MA) appeared, indicating several model prediction candidates: ARIMA(1,1,0), ARIMA(2,1,0), ARIMA(0,1,2), ARIMA(0,1,3), ARIMA(2,1,2), ARIMA(2,1,1). Through model selection based on residual analysis and model selection criteria (AIC and BIC), the best-fitted model is ARIMA(2,1,1), otherwise known as a log-differenced ARMA(2,1) model.

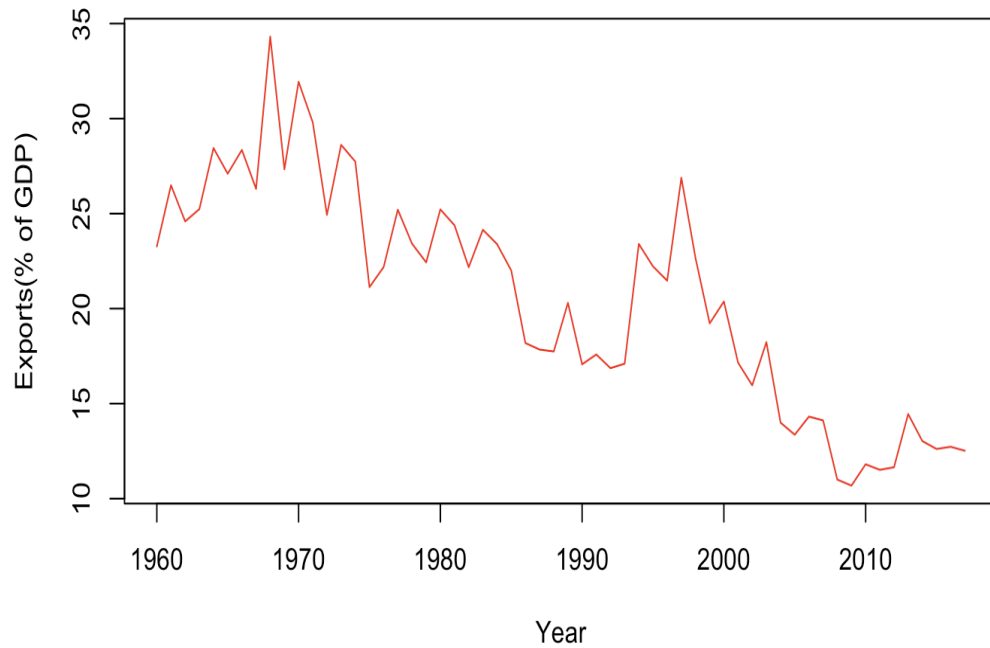
$$y''_t = -0.3981y_{t-1} - 0.3003y_{t-2} - w_{t-1} + w_t$$

Thus, we performed forecasting on the ARIMA(2,1,1) model, which demonstrated that while historical data from 1960 to 2017 displayed volatility, there can be a relatively stable trend post-2017. In other words, the forecasting visualization illustrated trends of stability post-2020, but historical volatility and the widening confidence intervals ranging from -0.27 to 0.24 log-differenced exports emphasizes the possibility of unpredictable shocks and future uncertainties. Overall, our analysis gave a better understanding of how export performance has changed through the span of 1960 to post-2020, while giving us predictions on how it might change in the future.

Plots

Model 1.1

Model 1.1: Central African Republic Exports



Mode 1.2

Model 1.2: C.A.R. Exports with Kernel Smoothing

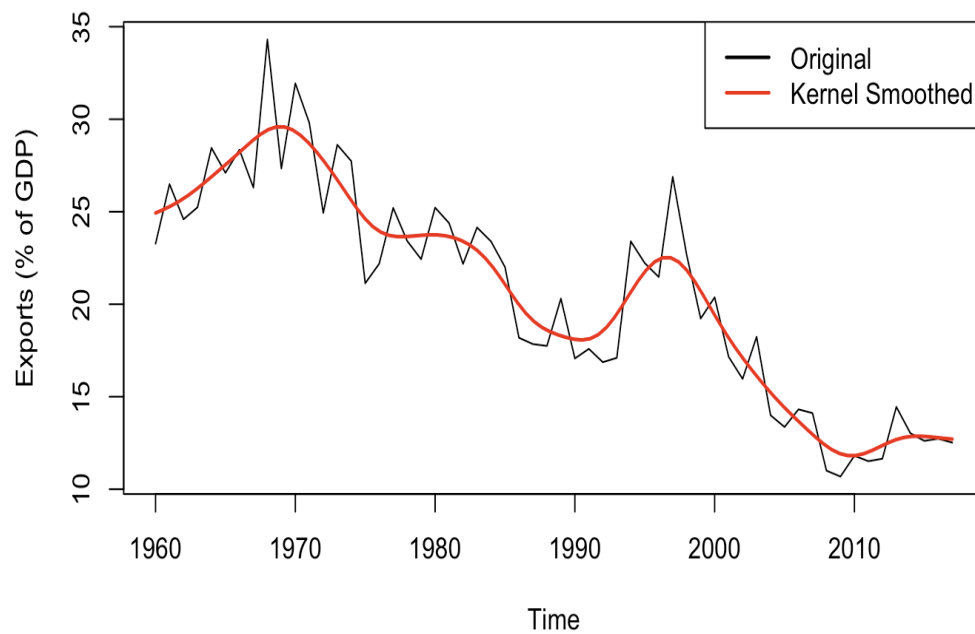


Figure 1.1: ACF Plot for Raw Data

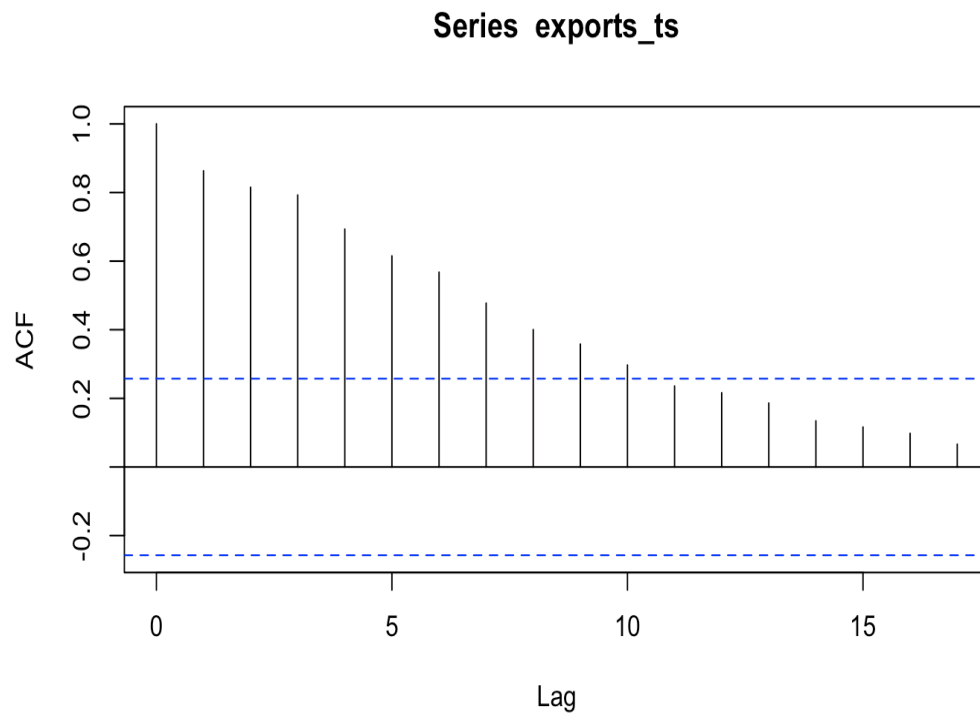
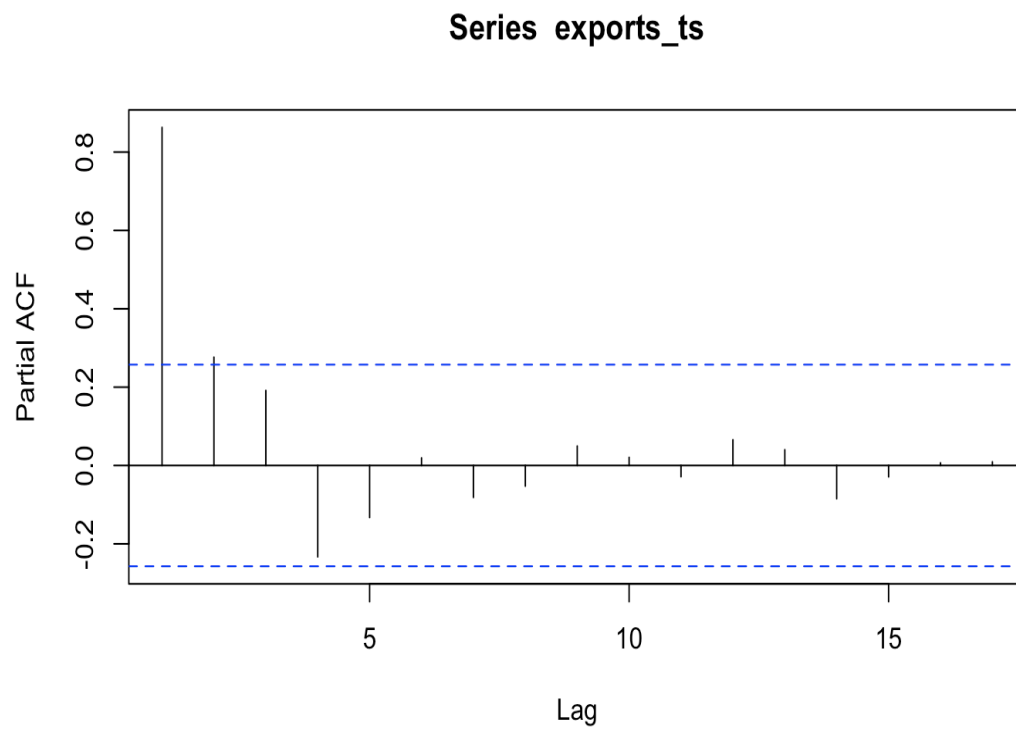


Figure 1.2: PACF Plot for Raw Data



Model 2.1

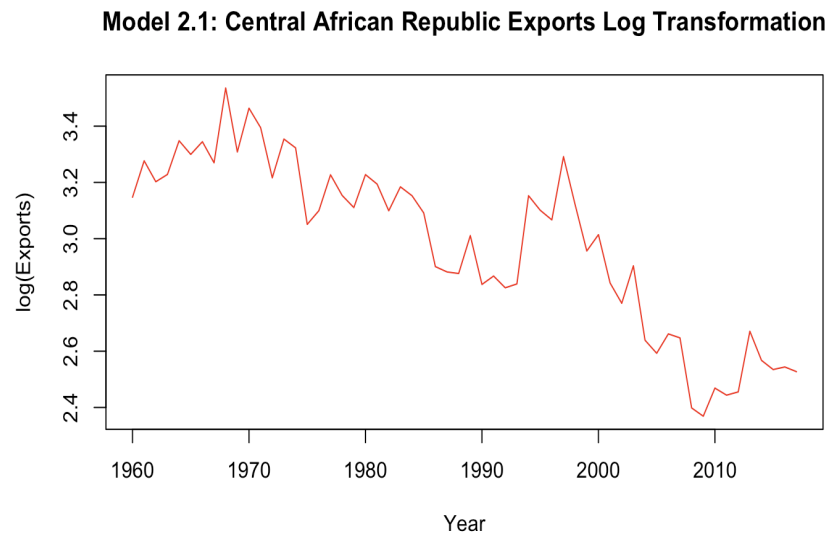


Figure 2.1: ACF for Log Transformed Model

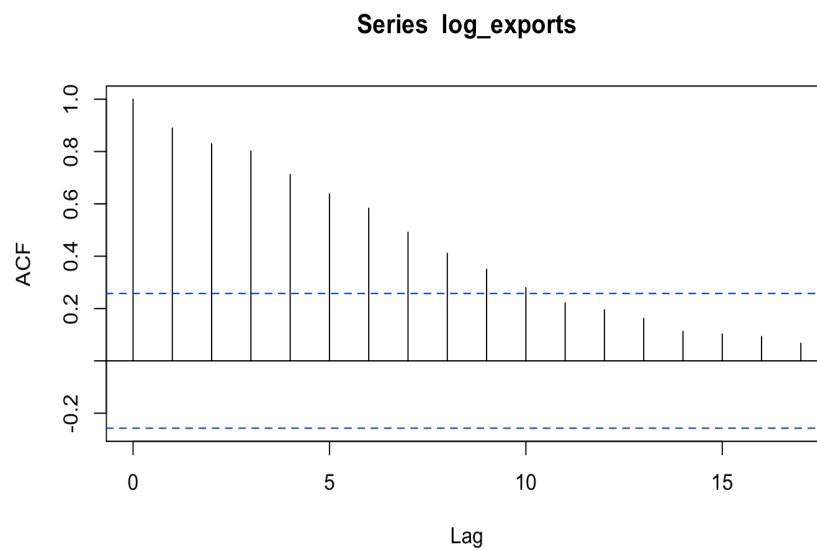
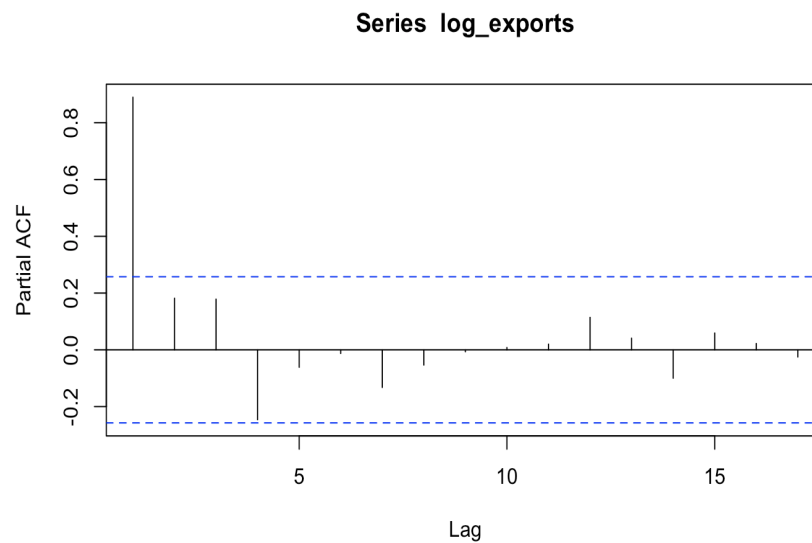


Figure 2.2: PACF Plot for Log Transformed Model



Model 3.1

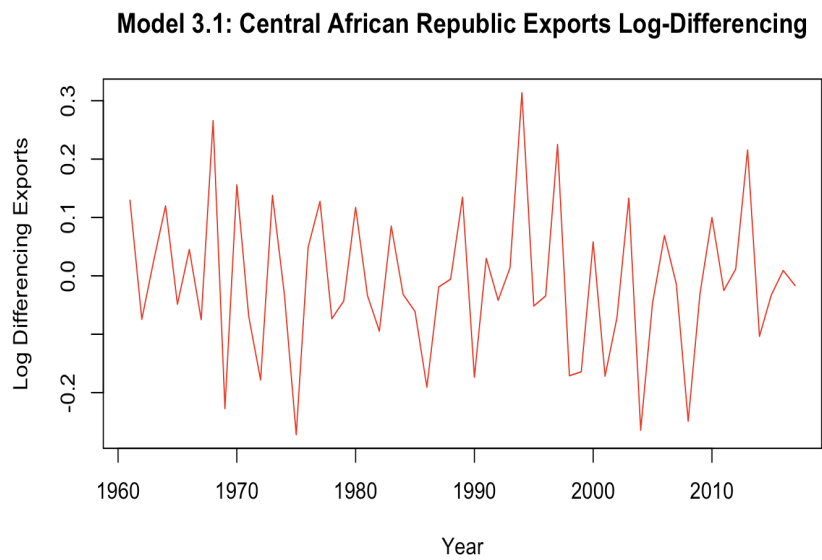


Figure 3.1: ACF for Log Differencing Model

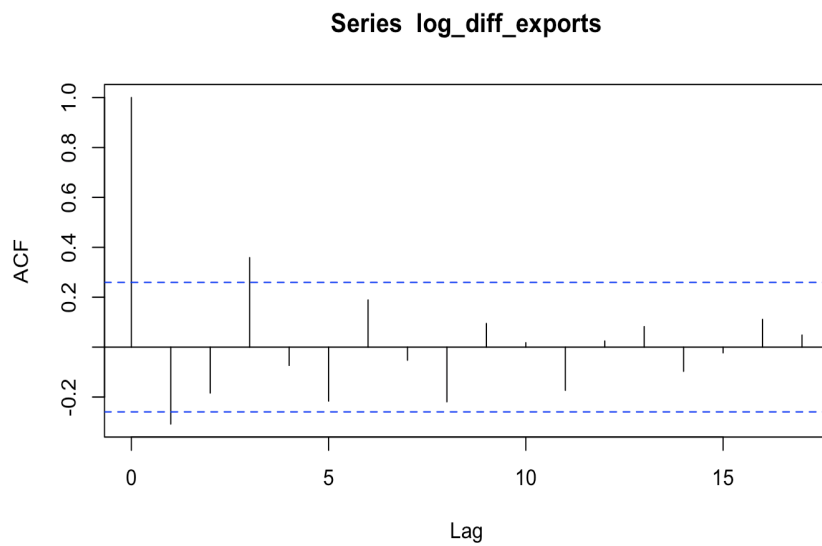
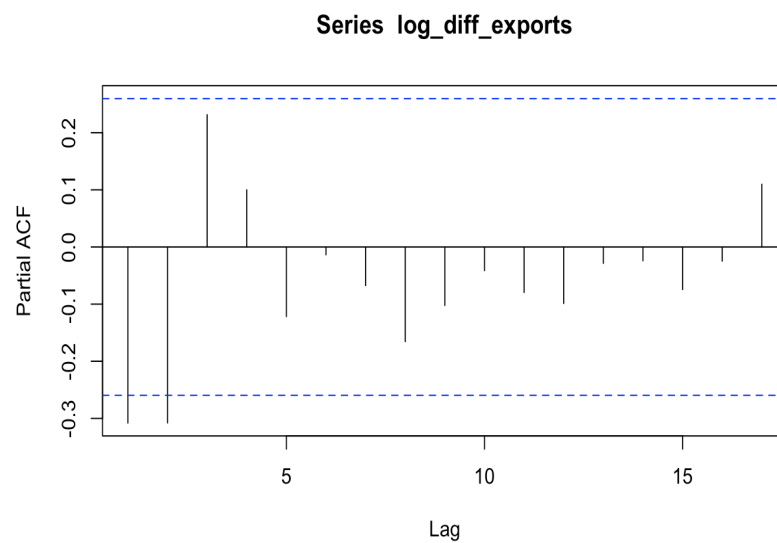
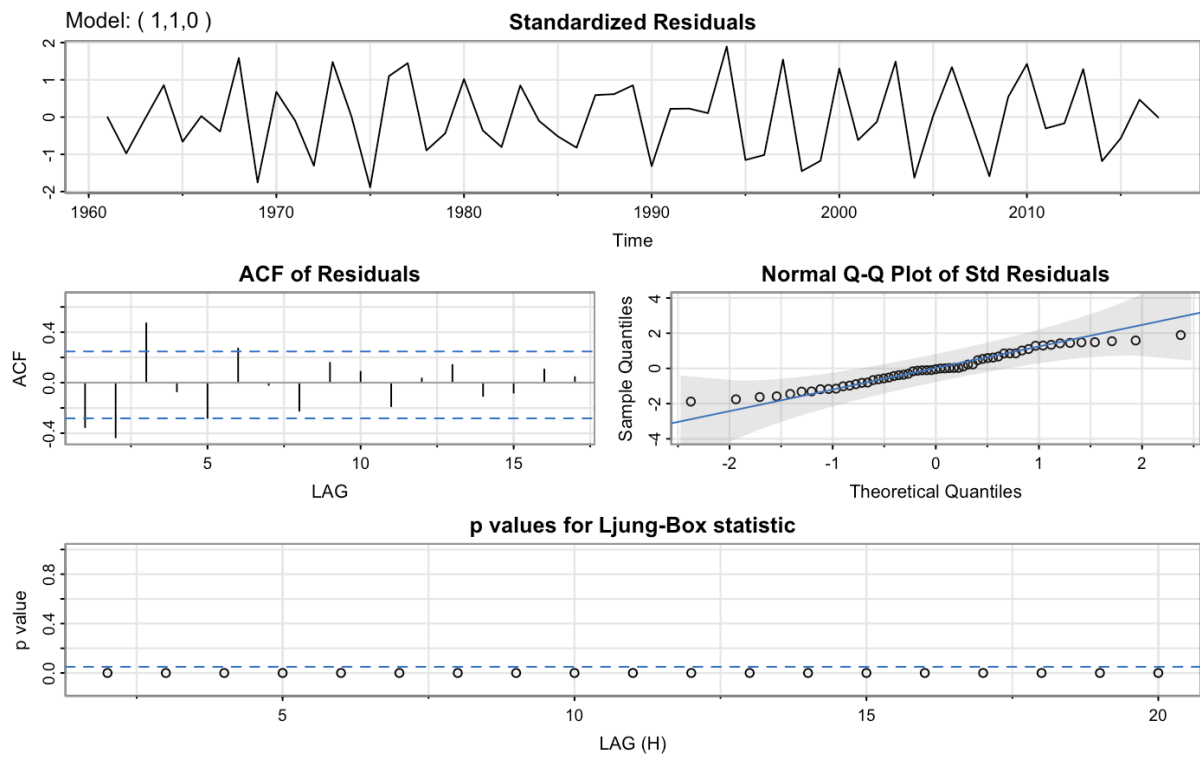


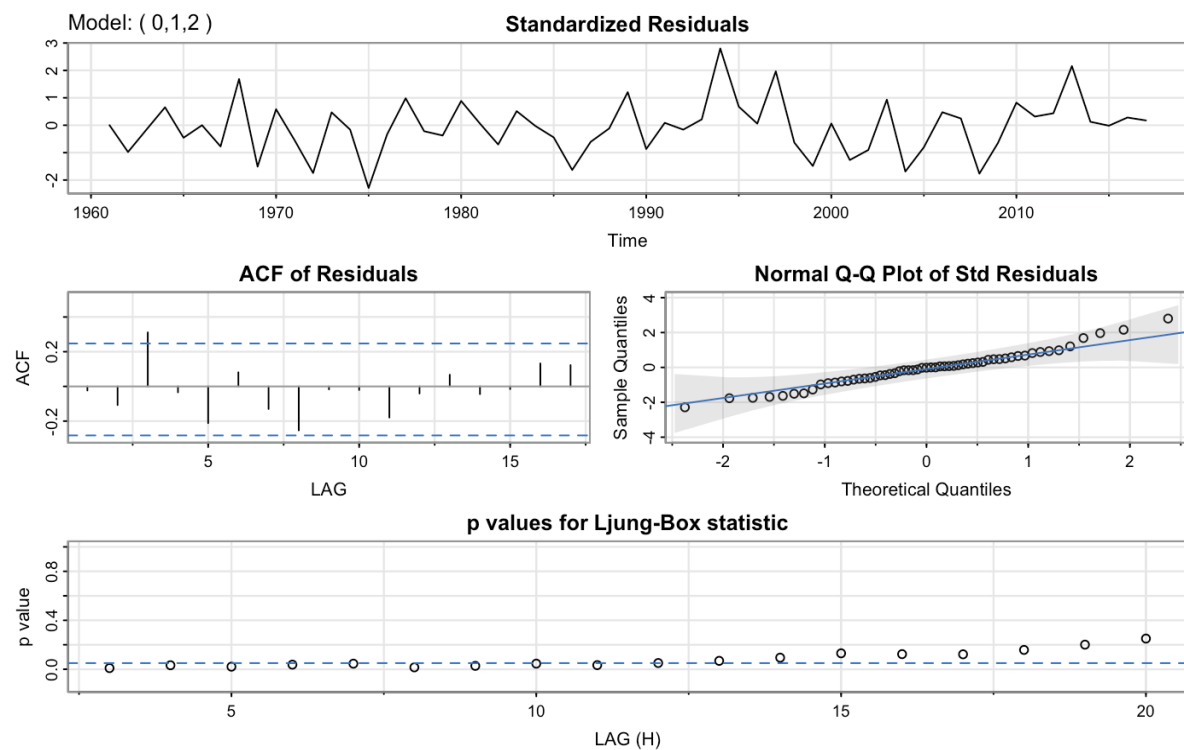
Figure 3.2: ACF for Log Differencing Model



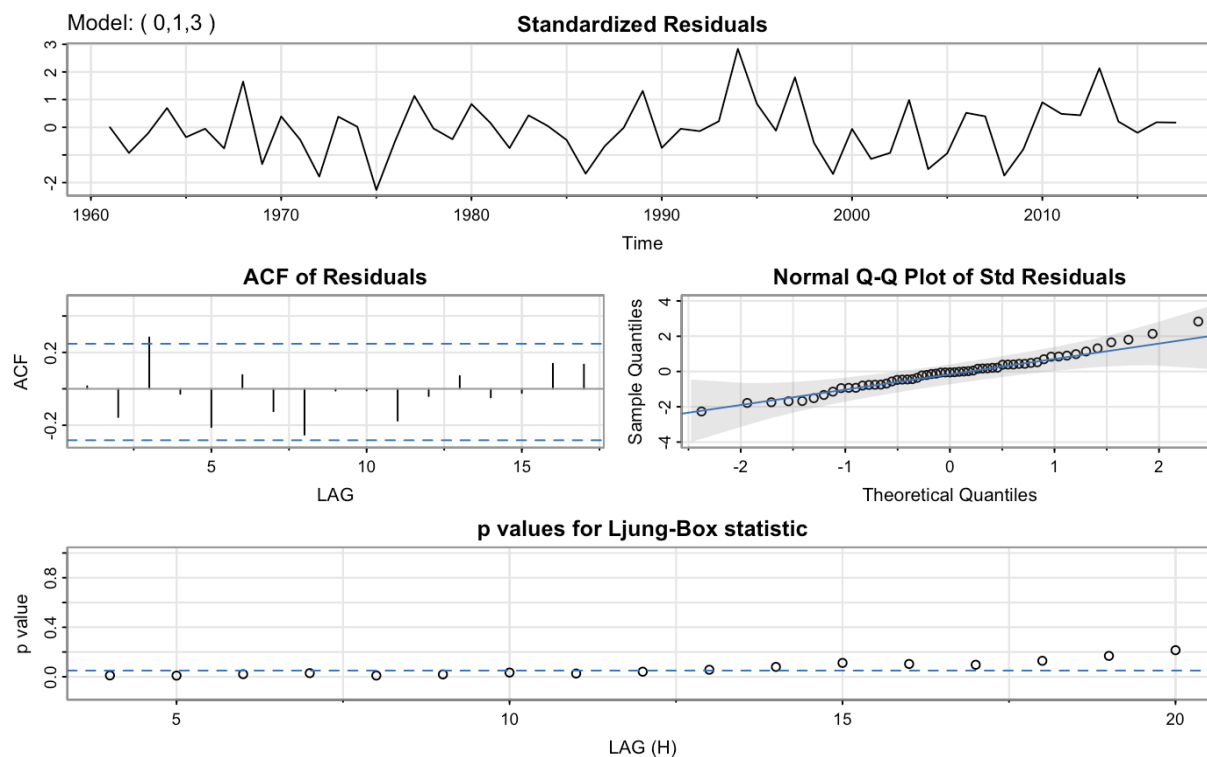
Model 4.1



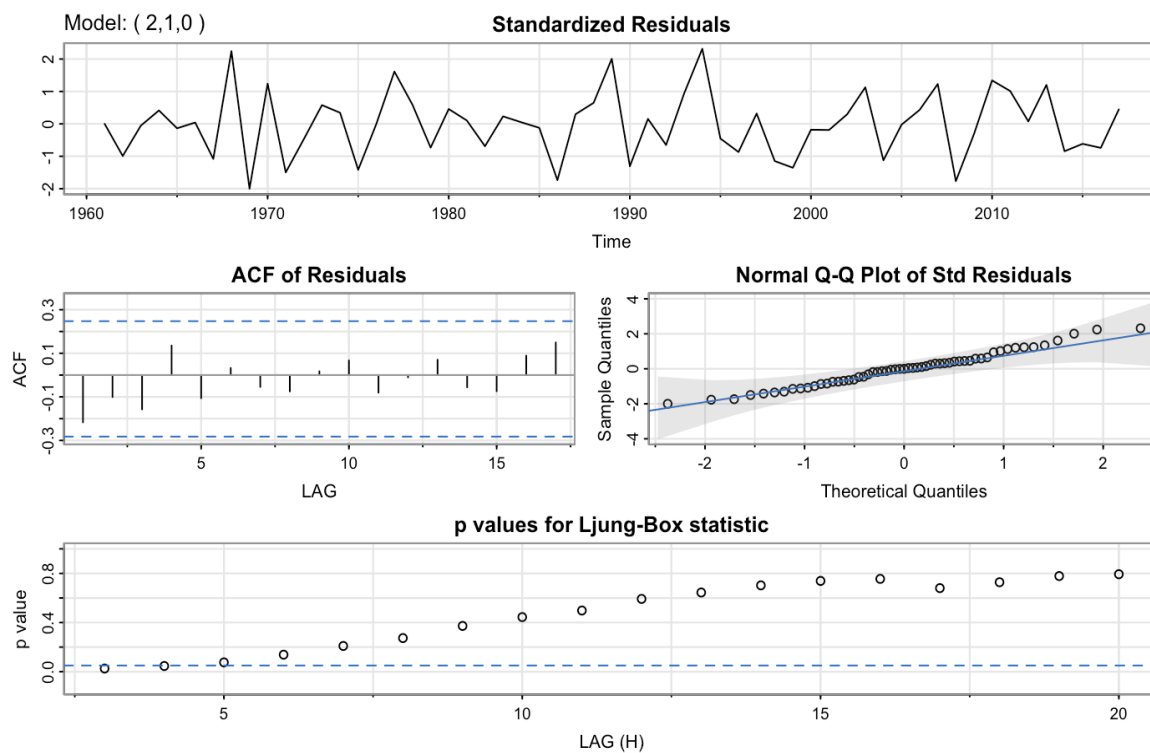
Model 5.1



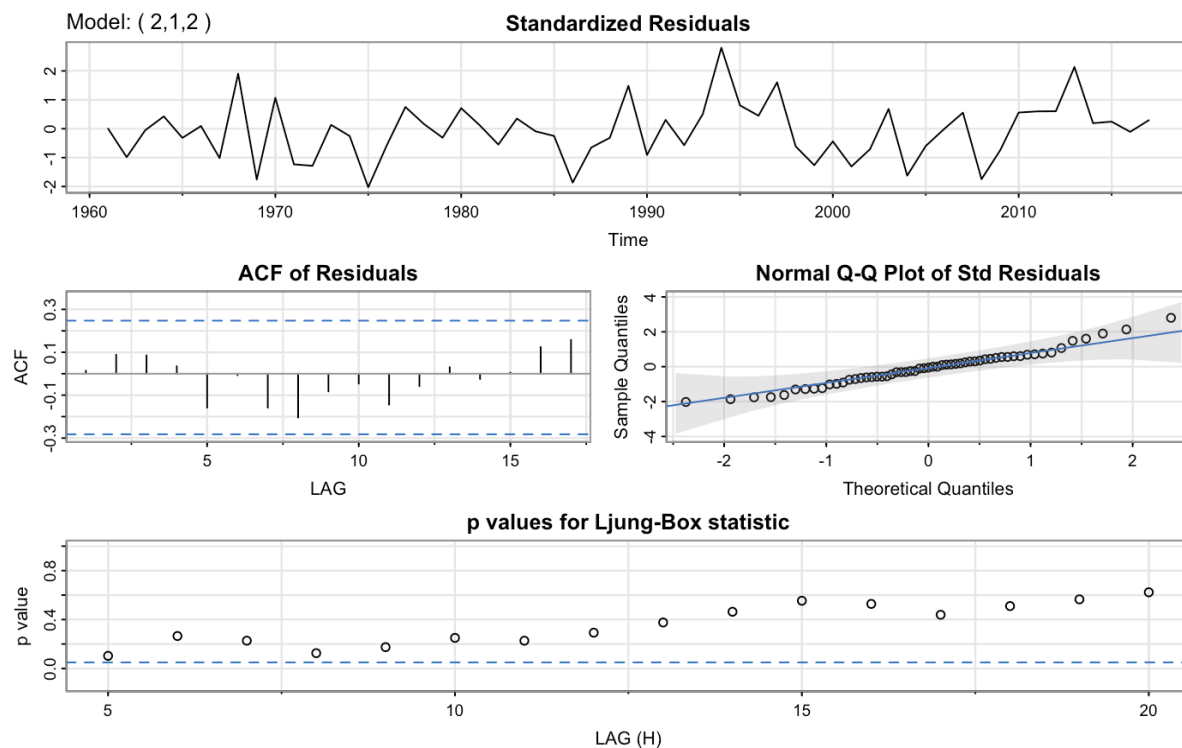
Model 6.1



Model 7.1



Model 8.1



Model 9.1

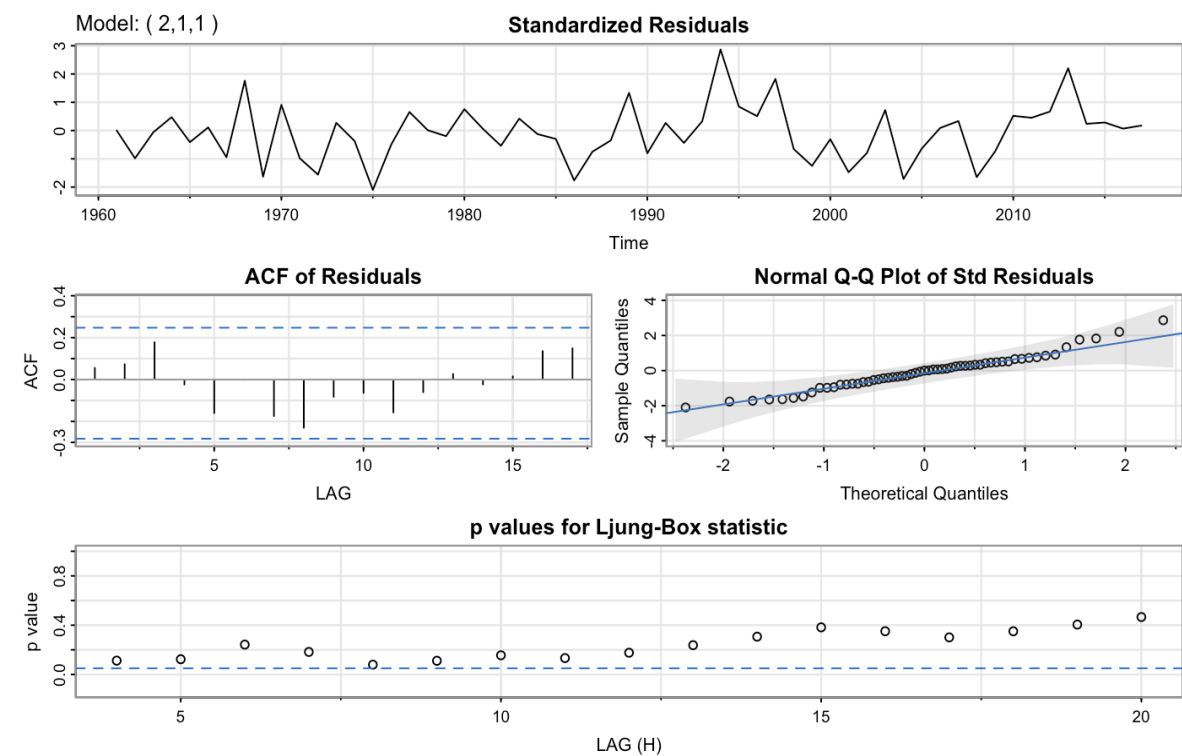


Figure 9.1

